

Unified Military Analytics and Comparison Dashboard-DV

Final Report

INTERNSHIP OF
DATA VISUALIZATION
IN
INFOSYS SPRINGBOARD

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ABSTRACT

This project aims to develop a fully interactive dashboard suite for analyzing global military power in 2025. Using Python, military metrics will be scraped from GlobalFirepower.com for 140+ countries via a pre-defined list (links_for_military_data.txt).

The project leverages Python-based data pipelines to scrape, clean, and process open-source defense data from GlobalFirepower.com. Unlike traditional single-platform dashboards, GDM emphasizes cross-platform flexibility — allowing deployment in tools like Tableau, Power BI, or Python-based web apps built using Streamlit or Dash.

Dashboards planned:

- Quick Stats – A global overview of rankings, trends, and highlights.
- Nation Overview – Detailed analysis of individual country capabilities.
- Compare Powers – Side-by-side military and economic comparison.
- Coalition Builder – Interactive simulation of alliance strength and combined assets.

The final deliverable is a comprehensive analytics suite that enables users to compare nations, evaluate coalitions, and analyze KPIs like Power Index Rank Gap, Assets per Capita, and Defense Budget-to-GDP Ratio through intuitive and dynamic visualizations.

Milestone 1: Data Collection and Preparation

Module 1: Scraping Setup and Execution

1.1 Methodologies:

1.1.1 Development Environment:

Google Collab: cloud based platform that can be used for python development, without need of any local installation.

1.1.2 Data Source:

File provided: links_for_military_data.txt

So this file has various links of GlobalFirepower, which is a online platform that provides data related to various countries military strength.

So the urls are:

```
base_url:'https://www.globalfirepower.com/countries-listing.php'
other_sources: {
    'https://www.globalfirepower.com/total-population-by-country.php':
'total_population',
    'https://www.globalfirepower.com/available-military-manpower.php':
'total_military_manpower',
    'https://www.globalfirepower.com/manpower-fit-for-military-
service.php': 'fit_for_service',
    'https://www.globalfirepower.com/manpower-reaching-military-age-
annually.php': 'population_reaching_military_age_annually',
    'https://www.globalfirepower.com/active-military-manpower.php':
'active_personnel',
    'https://www.globalfirepower.com/active-reserve-military-
manpower.php': 'reserve_personnel',
    'https://www.globalfirepower.com/manpower-paramilitary.php':
'paramilitary',
```

['https://www.globalfirepower.com/aircraft-total.php':](https://www.globalfirepower.com/aircraft-total.php)
 'total_military_aircraft',
['https://www.globalfirepower.com/aircraft-total-fighters.php':](https://www.globalfirepower.com/aircraft-total-fighters.php)
 'fighter_aircraft',
['https://www.globalfirepower.com/aircraft-total-attack-types.php':](https://www.globalfirepower.com/aircraft-total-attack-types.php)
 'attack_aircraft',
['https://www.globalfirepower.com/aircraft-total-transport.php':](https://www.globalfirepower.com/aircraft-total-transport.php)
 'transport_aircraft',
['https://www.globalfirepower.com/aircraft-total-trainers.php':](https://www.globalfirepower.com/aircraft-total-trainers.php)
 'trainer_aircraft',
['https://www.globalfirepower.com/aircraft-total-special-mission.php':](https://www.globalfirepower.com/aircraft-total-special-mission.php)
 'special_mission_aircraft',
['https://www.globalfirepower.com/aircraft-total-tanker-fleet.php':](https://www.globalfirepower.com/aircraft-total-tanker-fleet.php)
 'tanker_aircraft',
['https://www.globalfirepower.com/aircraft-helicopters-total.php':](https://www.globalfirepower.com/aircraft-helicopters-total.php)
 'total_military_helicopters',
['https://www.globalfirepower.com/aircraft-helicopters-attack.php':](https://www.globalfirepower.com/aircraft-helicopters-attack.php)
 'attack_helicopters',
['https://www.globalfirepower.com/armor-tanks-total.php':](https://www.globalfirepower.com/armor-tanks-total.php) 'tanks',
['https://www.globalfirepower.com/armor-apc-total.php':](https://www.globalfirepower.com/armor-apc-total.php)
 'armored_fighting_vehicles',
['https://www.globalfirepower.com/armor-self-propelled-guns-total.php':](https://www.globalfirepower.com/armor-self-propelled-guns-total.php) 'self_propelled_artillery',
['https://www.globalfirepower.com/armor-towed-artillery-total.php':](https://www.globalfirepower.com/armor-towed-artillery-total.php)
 'towed_artillery',
['https://www.globalfirepower.com/armor-mrls-total.php':](https://www.globalfirepower.com/armor-mrls-total.php)
 'rocket_projectors',
['https://www.globalfirepower.com/navy-ships.php':](https://www.globalfirepower.com/navy-ships.php) 'total_naval_fleet',
['https://www.globalfirepower.com/navy-force-by-tonnage.php':](https://www.globalfirepower.com/navy-force-by-tonnage.php)
 'total_naval_fleet_tonnage_mt',
['https://www.globalfirepower.com/navy-aircraft-carriers.php':](https://www.globalfirepower.com/navy-aircraft-carriers.php)
 'aircraft_carriers',
['https://www.globalfirepower.com/navy-helo-carriers.php':](https://www.globalfirepower.com/navy-helo-carriers.php)
 'helicopter_carriers',

<https://www.globalfirepower.com/navy-submarines.php>:
 'submarines',
<https://www.globalfirepower.com/navy-destroyers.php>: 'destroyers',
<https://www.globalfirepower.com/navy-frigates.php>: 'frigates',
<https://www.globalfirepower.com/navy-corvettes.php>: 'corvettes',
<https://www.globalfirepower.com/navy-patrol-coastal-craft.php>:
 'coastal_patrol_craft',
<https://www.globalfirepower.com/navy-mine-warfare-craft.php>:
 'mine_warfare_craft',
<https://www.globalfirepower.com/defense-spending-budget.php>:
 'defense_budget_usd',
<https://www.globalfirepower.com/external-debt-by-country.php>:
 'external_debt_usd',
<https://www.globalfirepower.com/purchasing-power-parity.php>:
 'purchasing_power_parity_usd',
<https://www.globalfirepower.com/reserves-of-foreign-exchange-and-gold.php>: 'foreign_exchange_and_gold_reserves_usd',
<https://www.globalfirepower.com/major-serviceable-airports-by-country.php>: 'total_serviceable_airports',
<https://www.globalfirepower.com/labor-force-by-country.php>:
 'labour_force',
<https://www.globalfirepower.com/major-ports-and-terminals.php>:
 'major_ports_and_terminals',
<https://www.globalfirepower.com/merchant-marine-strength-by-country.php>: 'total_merchant_marine_fleet',
<https://www.globalfirepower.com/railway-coverage.php>:
 'railway_coverage_km',
<https://www.globalfirepower.com/roadway-coverage.php>:
 'roadway_coverage_km',
<https://www.globalfirepower.com/oil-production-by-country.php>:
 'oil_production_bbl',
<https://www.globalfirepower.com/oil-consumption-by-country.php>:
 'oil_consumption_bbl',
<https://www.globalfirepower.com/proven-oil-reserves-by-country.php>: 'proven_oil_reserves_bbl',

```

'https://www.globalfirepower.com/natural-gas-production-by-
country.php': 'natural_gas_production_cum',
'https://www.globalfirepower.com/natural-gas-consumption-by-
country.php': 'natural_gas_consumption_cum',
'https://www.globalfirepower.com/proven-natural-gas-reserves-by-
country.php': 'proven_natural_gas_reserves_cum',
'https://www.globalfirepower.com/coal-production-by-country.php':
'coal_production_cum',
'https://www.globalfirepower.com/coal-consumption-by-country.php':
'coal_consumption_mt',
'https://www.globalfirepower.com/proven-coal-reserves-by-
country.php': 'proven_coal_reserves_cum',
'https://www.globalfirepower.com/square-land-area.php':
'total_land_area_sq_km',
'https://www.globalfirepower.com/coastline-coverage.php':
'coastline_coverage_km',
'https://www.globalfirepower.com/border-coverage.php':
'border_coverage_km',
'https://www.globalfirepower.com/waterway-coverage.php':
'waterway_coverage_km'
}

```

1.1.3 Tools and libraries:

The tools and libraries used for completing this milestone are:

- i. **Requests:** used to send the Http requests and also to fetch the content from the web page.
- ii. **BeautifulSoup (bs4):** used for parsing and extracting the HTML elements
- iii. **pandas:** used for structuring and storing the scraped data.
- iv. **time:** used for polite scraping i.e. to introduce delays between the requests.

1.2 Implementation steps:

i. Import all necessary libraries:

Here for this module import Beautiful Soup, pandas, requests and time. If not installed then, run pip install.

ii. Define the Data Source URLs:

From the given list of URLs provided in the file **links_for_military.txt**, copy all the URLs and metrics and create a dictionary of that and store it inside variable **URLS** in the format where key represents a URL and the corresponding value represents the metric name.

iii. Then set the request headers:

Define a browser-like user agent and helps avoid request blocking by the website.

iv. Create a empty data structure named **all_data:**

To store the data i.e the country name and its respective metric values extracted from the website

v. Next send the requests to the web pages:

In this step access the urls one by one and also ensure that the pages are loaded successfully.

vi. Parse the web page content:

With the help of beautiful soup library extract the html content from the response.

```
soup = BeautifulSoup(response.text, "html.parser")
```


After that for knowing the tag names, open any one of the urls, right click on whatever we want and select inspect, from there we can get div class name.

So the country name is inside - (div.longFormName)

For metric value:

if metric is pwr_index then its div.pwrIndxContainer.
else use div.valueContainer.

vii. Store the data:

Use the country name as the common reference key.

`All_data[country][metric]=value`

In case of new values, merge them with the existing country data.

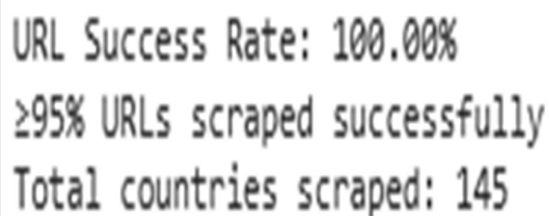
And handle the exceptions using try/except.

viii. Check whether all URLs are successfully scraped.

ix. Then convert the extracted data to DataFrame and save it as military_raw_data.csv

1.3 Outcomes:

- All URLs get scrapped successfully



```
URL Success Rate: 100.00%
≥95% URLs scraped successfully
Total countries scraped: 145
```

Figure 1: URL Success Rate

- No of rows and columns extracted:

```
[3]
✓ Os
print("No of rows:",df.shape[0])
print("No of columns:",df.shape[1])

... No of rows: 145
No of columns: 56
```

Figure 2: No. of Columns and Rows extracted

- Countries and metrics extracted:

```
print(df['Country'])

... 0 United States
1 Russia
2 China
3 India
4 South Korea
...
140 Kosovo
141 Somalia
142 Central African Republic
143 Benin
144 Bhutan
Name: Country, Length: 145, dtype: object

print(df.columns)

Index(['Country', 'pwr_index', 'total_population', 'total_military_manpower',
'fit_for_service', 'population_reaching_military_age_annually',
'active_personnel', 'reserve_personnel', 'paramilitary',
'total_military_aircraft', 'fighter_aircraft', 'attack_aircraft',
'transport_aircraft', 'trainer_aircraft', 'special_mission_aircraft',
'tanker_aircraft', 'total_military_helicopters', 'attack_helicopters',
'tanks', 'armored_fighting_vehicles', 'self_propelled_artillery',
'towed_artillery', 'rocket_projectors', 'total_naval_fleet',
'total_naval_fleet_tonnage_mt', 'aircraft_carriers',
'helicopter_carriers', 'submarines', 'destroyers', 'frigates',
'corvettes', 'coastal_patrol_craft', 'mine_warfare_craft',
'defense_budget_usd', 'external_debt_usd',
'purchasing_power_parity_usd', 'foreign_exchange_and_gold_reserves_usd',
'total_serviceable_airports', 'labour_force',
'major_ports_and_terminals', 'total_merchant_marine_fleet',
'railway_coverage_km', 'roadway_coverage_km', 'oil_production_bbl',
'oil_consumption_bbl', 'proven_oil_reserves_bbl',
'natural_gas_production_cum', 'natural_gas_consumption_cum',
'proven_natural_gas_reserves_cum', 'coal_production_cum',
'coal_consumption_mt', 'proven_coal_reserves_cum',
'total_land_area_sq_km', 'coastline_coverage_km', 'border_coverage_km',
'waterway_coverage_km'],
dtype='object')
```

Figure 3: countries and metrics extracted

- military_raw_data.csv

	A	B	C	D	E	F	G	H	I	J	K
1	Country	pwr_index	total_population	total_military_manpower	fit_for_service	population_reaching_military_age_annually	active_personnel	reserve_personnel	paramilitary	total_military_aircraft	fighter_aircraft at
2	United States	PwrIndex: 0.0744	34,19,63,408	15,04,63,900	12,48,16,644	44,45,524	13,28,000	7,99,500	0	13,043	1,790
3	Russia	PwrIndex: 0.0788	14,08,20,810	6,90,02,197	4,61,89,226	12,67,387	13,20,000	20,00,000	2,50,000	4,292	833
4	China	PwrIndex: 0.0788	1,41,50,43,270	76,41,23,366	62,68,64,169	1,98,10,606	20,35,000	5,10,000	6,25,000	3,309	1,212
5	India	PwrIndex: 0.1184	1,40,91,28,296	66,22,90,299	52,27,86,598	2,39,55,181	14,55,550	11,55,000	25,27,000	2,229	513
6	South Korea	PwrIndex: 0.1656	5,20,81,799	2,60,40,900	2,13,53,538	4,16,654	6,00,000	31,00,000	1,20,000	1,592	315
7	United Kingdom	PwrIndex: 0.1785	6,94,50,055	3,14,91,165	2,51,92,932	7,53,050	1,84,800	9,24,000	0	631	113
8	France	PwrIndex: 0.1878	6,83,74,591	3,00,84,820	2,37,94,358	7,52,121	2,00,000	26,000	1,50,000	976	226
9	Japan	PwrIndex: 0.1839	12,32,01,945	5,29,76,836	4,28,74,277	11,08,818	2,47,150	56,000	25,000	1,443	217
10	Turkiye	PwrIndex: 0.1902	8,41,19,531	4,29,85,080	3,60,87,279	14,30,032	3,55,200	3,78,700	1,50,000	1,083	201
11	Italy	PwrIndex: 0.2164	6,09,64,931	2,74,34,219	2,21,91,235	5,48,684	1,65,500	18,500	1,05,000	729	89
12	Brazil	PwrIndex: 0.2415	22,00,51,512	11,22,26,271	8,86,80,759	35,20,824	3,60,000	3,40,000	2,00,000	513	43
13	Pakistan	PwrIndex: 0.2513	25,23,63,571	10,85,16,336	8,58,03,614	47,94,908	6,54,000	5,50,000	5,00,000	1,399	328
14	Indonesia	PwrIndex: 0.2557	28,15,62,465	13,79,65,608	11,45,95,223	47,86,562	4,90,000	4,00,000	2,50,000	459	41
15	Germany	PwrIndex: 0.2601	8,41,19,100	3,86,94,786	3,09,55,829	8,41,191	1,81,600	34,000	0	584	129
16	Israel	PwrIndex: 0.2661	94,02,617	39,49,099	32,81,513	1,31,637	1,70,000	4,65,000	35,000	611	240
17	Iran	PwrIndex: 0.3048	8,83,86,937	4,94,96,685	4,15,41,860	14,14,191	6,10,000	3,50,000	2,20,000	551	188
18	Spain	PwrIndex: 0.3242	4,72,80,433	2,17,48,999	1,76,35,602	3,30,963	1,33,282	15,150	78,470	461	137
19	Australia	PwrIndex: 0.3298	2,67,68,598	1,12,42,811	92,08,398	2,94,455	57,350	32,050	0	327	24
20	Egypt	PwrIndex: 0.3427	11,12,47,248	4,56,11,372	3,82,69,053	16,68,709	4,40,000	4,80,000	3,00,000	1,093	238
21	Ukraine	PwrIndex: 0.3755	3,56,61,826	1,81,87,531	1,27,31,772	3,92,280	9,90,000	12,00,000	1,00,000	324	70
22	Poland	PwrIndex: 0.3776	3,87,46,310	1,89,85,692	1,55,76,017	4,64,956	2,02,100	3,50,000	50,000	479	59

Figure 4: Snapshot of military_raw_data.csv

Module 2: Data Cleaning and Structuring:

2.1 Methodologies:

2.1.1 Development Environment:

Google Collab: cloud based platform that can be used for python development, without need of any local installation.

2.1.2 Data Source:

military_raw_data.csv

2.1.3 Tools and Libraries:

pandas: Used for loading the dataset, cleaning column values, handling missing data, and saving the cleaned file.

re (Regular Expressions): Used to remove unwanted characters such as commas, percentage signs, and special symbols from numeric values.

2.2 Steps for Data Cleaning:

- i. Import all the necessary libraries.
- ii. Then load the military_raw_data.csv
- iii. Check the data types of the columns:

```
print(df.dtypes)

--- Country                                object
    pwr_index                             object
    total_population                       object
    total_military_manpower               object
    fit_for_service                       object
    population_reaching_military_age_annually object
    active_personnel                     object
    reserve_personnel                    object
    paramilitary                         object
    total_military_aircraft              object
    fighter_aircraft                     object
    attack_aircraft                      int64
    transport_aircraft                   int64
    trainer_aircraft                     object
    special_mission_aircraft             int64
    tanker_aircraft                      int64
    total_military_helicopters            object
    attack_helicopters                   object
    tanks                                object
    armored_fighting_vehicles             object
    self_propelled_artillery              object
    towed_artillery                       object
    rocket_projectors                     object
    total_naval_fleet                     int64
    total_naval_fleet_tonnage_mt           object
    aircraft_carriers                     int64
    helicopter_carriers                   int64
    submarines                           int64
    destroyers                            int64
    frigates                             int64
    corvettes                             int64
    coastal_patrol_craft                  int64
    mine_warfare_craft                   int64
    defense_budget_usd                    object
    external_debt_usd                     object
    purchasing_power_parity_usd           object
    foreign_exchange_and_gold_reserves_usd object
    total_serviceable_airports            object
    labour_force                          object
    major_ports_and_terminals             int64
    total_merchant_marine_fleet            object
    railway_coverage_km                    object
    roadway_coverage_km                   object
    oil_production_bbl                     object
    oil_consumption_bbl                   object
    proven_oil_reserves_bbl                object
    natural_gas_production_cum             object
    natural_gas_consumption_cum            object
    proven_natural_gas_reserves_cum        object
    coal_production_cum                   object
    coal_consumption_mt                    object
    proven_coal_reserves_cum               object
    total_land_area_sq_km                 object
    coastline_coverage_km                  object
    border_coverage_km                     object
    waterway_coverage_km                   object
    dtype: object
```

Figure 5: data types using df.dtypes

- iv. Its visible that there is a need to format the columns.
- v. A custom function `clean_value(val)` is created to: Handle missing values, Convert all values into string format, Remove commas, percentage symbols, and extra characters, Retain only numeric values, Convert cleaned values into numeric format.
- vi. **Apply Cleaning Across Columns:** The cleaning function is applied to all columns except the country name, Ensures consistency across all numeric indicators.
- vii. **Standardize Column Names:** Column names are converted into a consistent format: Lowercase, Underscore-separated, Special characters removed, Improves readability and compatibility with analysis tools.
- viii. **Identify Missing Values:** Missing values are counted for each column. Columns containing missing data are identified for validation and review.

```
[25]
✓ Os missing_metrics = df.isna().sum()
print("\nMissing metric counts per column:")
print(missing_metrics)

***
Missing metric counts per column:
Country                0
per_index              0
total_population       0
total_military_manpower 0
fit_for_service        0
population_reaching_military_age_annually 0
active_personnel       0
reserve_personnel      0
paramilitary           0
total_military_aircraft 0
fighter_aircraft       0
attack_aircraft        0
transport_aircraft     0
trainer_aircraft       0
special_mission_aircraft 0
tanker_aircraft        0
total_military_helicopters 0
attack_helicopters     0
tanks                 0
armored_fighting_vehicles 0
self_propelled_artillery 0
towed_artillery        0
rocket_projectors      0
total_naval_fleet      0
total_naval_fleet_tonnage_mt 98
aircraft_carriers      0
helicopter_carriers    0
submarines             0
destroyers             0
frigates              0
corvettes             0
coastal_patrol_craft   0
mine_warfare_craft     0
defense_budget_usd     0
external_debt_usd      0
purchasing_power_parity_usd 0
foreign_exchange_and_gold_reserves_usd 0
total_serviceable_airports 0
labour_force           0
major_ports_and_terminals 0
total_merchant_marine_fleet 0
railway_coverage_km    0
roadway_coverage_km    0
oil_production_bbl     0
oil_consumption_bbl    0
proven_oil_reserves_bbl 0
natural_gas_production_cum 0
natural_gas_consumption_cum 0
proven_natural_gas_reserves_cum 0
coal_production_cum    0
coal_consumption_mt    0
proven_coal_reserves_cum 0
total_land_area_sq_km  0
coastline_coverage_km  0
border_coverage_km     0
waterway_coverage_km   0
```

Figure 6: Checking the columns with missing values

So the column `total_naval_fleet_tonnage_mt` has the missing values. Replace it with 0 for better analysis.

ix. Convert Columns to Integer Where Needed:

Columns representing counts like `total_population`, `total_military_manpower`, `tanks`, `submarines`, etc., were converted to integer using `astype(int)`. Other columns like `budgets`, `tonnage`, or `power index` were kept as float

x. Then save the cleaned dataset: `military_cleaned.csv`

2.3 Outcomes:

So after data cleaning:

```
[32]
✓ Os
print(df.dtypes)

Country                object
pwr_index              float64
total_population       int64
total_military_manpower int64
fit_for_service        int64
population_reaching_military_age_annually int64
active_personnel       int64
reserve_personnel      int64
paramilitary           int64
total_military_aircraft int64
fighter_aircraft       int64
attack_aircraft        int64
transport_aircraft     int64
trainer_aircraft       int64
special_mission_aircraft int64
tanker_aircraft        int64
total_military_helicopters int64
attack_helicopters     int64
tanks                 int64
armored_fighting_vehicles int64
self_propelled_artillery int64
towed_artillery       int64
rocket_projectors      int64
total_naval_fleet      int64
total_naval_fleet_tonnage_mt float64
aircraft_carriers      int64
helicopter_carriers    int64
submarines             int64
destroyers             int64
frigates              int64
corvettes             int64
coastal_patrol_craft   int64
mine_warfare_craft     int64
defense_budget_usd     float64
external_debt_usd      float64
purchasing_power_parity_usd float64
foreign_exchange_and_gold_reserves_usd float64
total_serviceable_airports int64
labour_force           float64
major_ports_and_terminals int64
total_merchant_marine_fleet int64
railway_coverage_km    float64
roadway_coverage_km    float64
oil_production_bbl      float64
oil_consumption_bbl     float64
proven_oil_reserves_bbl float64
natural_gas_production_cum float64
natural_gas_consumption_cum float64
proven_natural_gas_reserves_cum float64
coal_production_cum     float64
coal_consumption_mt     float64
proven_coal_reserves_cum float64
total_land_area_sq_km   float64
coastline_coverage_km   float64
border_coverage_km      float64
waterway_coverage_km    float64
dtype: object
```

Figure 7: Data types after cleaning the data.

```
# Replace all missing values with 0
df.fillna(0, inplace=True)

# Verify missing values after replacement
print("Missing values after replacement:")
print(df.isna().sum())
```

... Missing values after replacement:

Country	0
pwr_index	0
total_population	0
total_military_manpower	0
fit_for_service	0
population_reaching_military_age_annually	0
active_personnel	0
reserve_personnel	0
paramilitary	0
total_military_aircraft	0
fighter_aircraft	0
attack_aircraft	0
transport_aircraft	0
trainer_aircraft	0
special_mission_aircraft	0
tanker_aircraft	0
total_military_helicopters	0
attack_helicopters	0
tanks	0
armored_fighting_vehicles	0
self_propelled_artillery	0
towed_artillery	0
rocket_projectors	0
total_naval_fleet	0
total_naval_fleet_tonnage_mt	0
aircraft_carriers	0
helicopter_carriers	0
submarines	0
destroyers	0
frigates	0
corvettes	0
coastal_patrol_craft	0
mine_warfare_craft	0
defense_budget_usd	0
external_debt_usd	0
purchasing_power_parity_usd	0
foreign_exchange_and_gold_reserves_usd	0
total_serviceable_airports	0
labour_force	0
major_ports_and_terminals	0
total_merchant_marine_fleet	0
railway_coverage_km	0
roadway_coverage_km	0
oil_production_bbl	0
oil_consumption_bbl	0
proven_oil_reserves_bbl	0
natural_gas_production_cum	0
natural_gas_consumption_cum	0
proven_natural_gas_reserves_cum	0
coal_production_cum	0
coal_consumption_mt	0
proven_coal_reserves_cum	0
total_land_area_sq_km	0
coastline_coverage_km	0
border_coverage_km	0
waterway_coverage_km	0
dtype:	int64

Figure 8: Missing Values after Replacement

	A	B	C	D	E	F	G	H	I	J	K	L
1	Country	pwr_index	total_population	total_military_manpower	fit_for_service	population_reaching_military_age_annually	active_personnel	reserve_personnel	paramilitary	total_military_aircraft	fighter_aircraft	attack
2	United States	0.0744	341963408	150463900	124816644	4445524	1328000	799500	0	13043	1790	
3	Russia	0.0788	140820810	69002197	46189226	1267387	1320000	2000000	250000	4292	833	
4	China	0.0788	1415043270	764123366	626864169	19810606	2035000	510000	625000	3309	1212	
5	India	0.1184	1409128296	662290299	522786598	23955181	1455550	1155000	2527000	2229	513	
6	South Korea	0.1656	52081799	26040900	21353538	416654	600000	3100000	120000	1592	315	
7	United Kingdom	0.1785	68459055	31491165	25192932	753050	184860	924000	0	631	113	
8	France	0.1878	68374591	30084820	23794358	752121	200000	26000	150000	976	226	
9	Japan	0.1839	123201945	52976836	42874277	1108818	247150	56000	25000	1443	217	
10	Turkiye	0.1902	84119531	42985080	36087279	1430032	355200	378700	150000	1083	201	
11	Italy	0.2164	60964931	27434219	22191235	548684	165500	18500	105000	729	89	
12	Brazil	0.2415	220051512	112226271	88680759	3520824	360000	340000	200000	513	43	
13	Pakistan	0.2513	252363571	108516336	85803614	4794908	654000	550000	500000	1399	328	
14	Indonesia	0.2557	281562465	137965608	114595923	4786562	400000	400000	250000	459	41	
15	Germany	0.2601	84119100	38694786	30955829	841191	181600	34000	0	584	129	
16	Israel	0.2661	9402617	3949099	3281513	131637	170000	465000	35000	611	240	
17	Iran	0.3048	88386937	49496685	41541860	1414191	610000	350000	220000	551	188	
18	Spain	0.3242	47280433	21748999	17635602	330963	133282	15150	78470	461	137	

Figure 9: Snapshot of military_cleaned.csv

Milestone 2: KPI Engineering and Dashboard Prep

Module 3: KPI Feature Engineering

3.1 Methodologies:

3.1.1 Development Environment:

Google Collab: cloud based platform that can be used for python development, without need of any local installation.

3.1.2 Data Source:

military_cleaned.csv

3.1.3 Tools and libraries:

The tools and libraries used for completing this milestone are:

- v. **Requests:** used to send the Http requests and also to fetch the content from the web page.
- vi. **BeautifulSoup (bs4):** used for parsing and extracting the HTML elements
- vii. **pandas:** used for structuring, storing the scraped data, Data cleaning, aggregation, and KPI calculations.
- viii. **NumPy:** Numerical and ratio-based computations

3.2 Implementation steps:

i. **Load Required Libraries**

- Import pandas and numpy for data processing.
- Import requests and BeautifulSoup for external metadata.

ii. **Load and Prepare Dataset**

- Load military_cleaned.csv dataset using Pandas.
- Standardize column names into lowercase.

iii. Calculate the KPI's:

1. Power Index Rank Gap:

- This metric will help to show how far a nation lags or leads compared to top military powers. So for this first calculate ranks of each country with the help of power index, where the one with lower power index has stronger military and also rank according to total population.
- Formula: $\text{power_index_rank_gap} = \text{population_rank} - \text{power_rank}$

2. Assets per Capita:

- This will help to measure the total assets available relative to the population size.
- Formula: $\text{assets_per_capita} = \frac{(\text{total_military_aircraft} + \text{tanks} + \text{total_naval_fleet} + \text{total_military_helicopters})}{\text{total_population}}$

3. budget_to_gdp_ratio:

- This shows how much of a country's economy (GDP) is spent on its military budget.
- Formula: $\text{budget_to_gdp_ratio} = \frac{\text{defense_budget_usd}}{\text{df["purchasing_power_parity_usd"]}}$

4. Budget per soldier:

- Indicates how much financial support each service member gets.
- Formula: $\text{budget_per_soldier} = \frac{\text{defense_budget_usd}}{\text{total_military_manpower}}$

5. air_power_density:

- Shows concentration of aircraft relative to population.
- Formula: $\text{air_power_density} = \frac{\text{total_military_aircraft}}{\text{total_population}}$

6. military_participation_ratio:

- This shows Reveals proportion of citizens engaged in active military service.
- Formula: $\text{military_participation_ratio} = \frac{\text{total_military_manpower}}{\text{total_population}}$

iii. Handle Infinite & Null Values

- Replace infinite values with NaN.
- Ensure clean numeric outputs for dashboards.

iv. Alliance (NATO) Feature Engineering

- Scrape NATO member countries from website
“<https://www.nato.int/en/about-us/organization/nato-member-countries>”, and store in nato_countries.
- Create binary flag is_nato (1 = NATO, 0 = Non-NATO).

v. Region & Continent Mapping

- Scrape country–region–continent mapping from
“<https://statisticstimes.com/geography/countries-by-continents.php>”
- Map region and continent to each country.
- Then Manually fix unmatched country names.

viii. Export Final Dataset

- Save engineered KPIs and metadata to final CSV.
- Save it as military_final.csv

3.3 Outcomes:

	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO
	power_rank	population_rank	power_index_rank_gap	assets_per_capita	budget_to_gdp_ratio	budget_per_soldier	air_power_density	military_participation_ratio	is_nato	region	continent
1	1	3	2	7.01E-05	0.036234818	5948.270648	2.81E-05	0.440000001	1	Northern America	North America
2	2	9	7	8.60E-05	0.021649485	1826.028815	3.05E-05	0.490000001	0	Eastern Europe	Europe
3	3	2	-1	8.32E-06	0.008557692	349.4200176	2.34E-06	0.54	0	Eastern Asia	Asia
4	4	2	-2	5.41E-06	0.005725191	113.2431921	1.58E-06	0.47	0	Southern Asia	Asia
5	5	28	23	9.34E-05	0.017671756	1777.922344	3.06E-05	0.500000001	0	Eastern Asia	Asia
6	6	21	15	1.80E-05	0.01932447	2270.495233	9.22E-06	0.459999996	1	Northern Europe	Europe
7	7	22	14	2.59E-05	0.01462766	1828.164503	1.43E-05	0.439999999	1	Western Europe	Europe
8	8	11	4	2.21E-05	0.009895833	1075.941946	1.17E-05	0.429999997	0	Eastern Asia	Asia
9	9	18	9	4.77E-05	0.015986395	1093.402641	1.29E-05	0.519999996	0	Western Asia	Asia
10	10	24	14	2.49E-05	0.00964516	1125.966079	1.20E-05	0.450000001	1	Southern Europe	Europe
11	11	7	-4	4.83E-06	0.006506791	233.0764425	2.33E-06	0.509999999	0	South America	South America
12	12	5	-7	1.79E-05	0.00569259	70.40414634	5.54E-06	0.430000002	0	Southern Asia	Asia
13	13	4	-9	4.74E-06	0.002710997	76.83074376	1.63E-06	0.490000001	0	South-eastern Asia	Asia
14	14	19	5	1.49E-05	0.009560229	1292.163756	6.94E-06	0.46	1	Western Europe	Europe
15	15	92	77	0.000225469	0.064755839	7723.280677	6.50E-05	0.419999985	0	Western Asia	Asia
16	16	17	1	2.83E-05	0.010729167	312.1421162	6.23E-06	0.560000003	0	Southern Asia	Asia
17	17	32	15	2.29E-05	0.010465089	1075.773648	9.75E-06	0.459999996	1	Southern Europe	Europe
18	18	54	36	1.86E-05	0.035253165	4954.277004	1.27E-05	0.419999994	0	Australia and New Zealand	Oceania
19	19	15	-4	4.68E-05	0.003078272	128.8042566	9.82E-06	0.410000003	0	Northern Africa	Africa
20	20	43	22	4.66E-05	0.095892857	2952.572287	9.09E-06	0.509999993	0	Eastern Europe	Europe
21	21	38	16	3.54E-05	0.030061728	2565.089542	1.24E-05	0.490000003	1	Eastern Europe	Europe

Figure 10: snapshot of military_final.csv

Module 4: Dashboard Planning and Prototyping

4.1 Tools Used:

- Figma – for designing low-fidelity wireframes and storyboard layouts
- Microsoft Power BI – for building the interactive operational dashboard using sample data

4.2 Dataset:

- military_final.csv

4.3 Implementation:

4.3.1 Figma wireframe for story board

Draft layout sketches or wireframes for:

o Quick Stats

o Nation Overview

o Compare Powers

o Coalition Builder

- In the figma, create the wireframe for the 4 dashboards.
- The navigation between the dashboard will happen with the help of the buttons.
- So sketch the buttons Quick Stats, Nation Overview, Compare Powers and Coalition Builder.

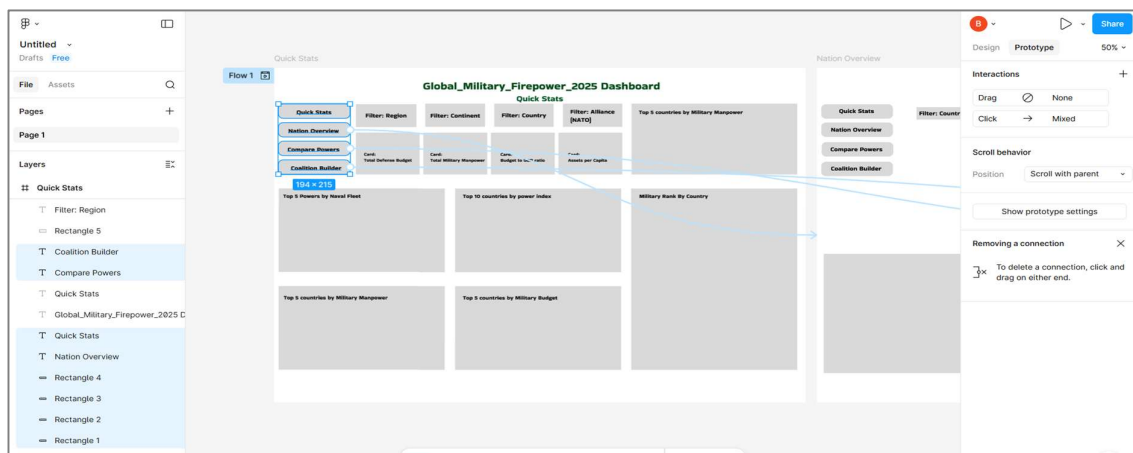


Figure 11. Wireframe for the dashboard.

- This image shows the flow of the dashboard from one to another when the button is clicked.
- Filters:

Set filters on the basis of Region, Continent, Country and Alliance (NATO).

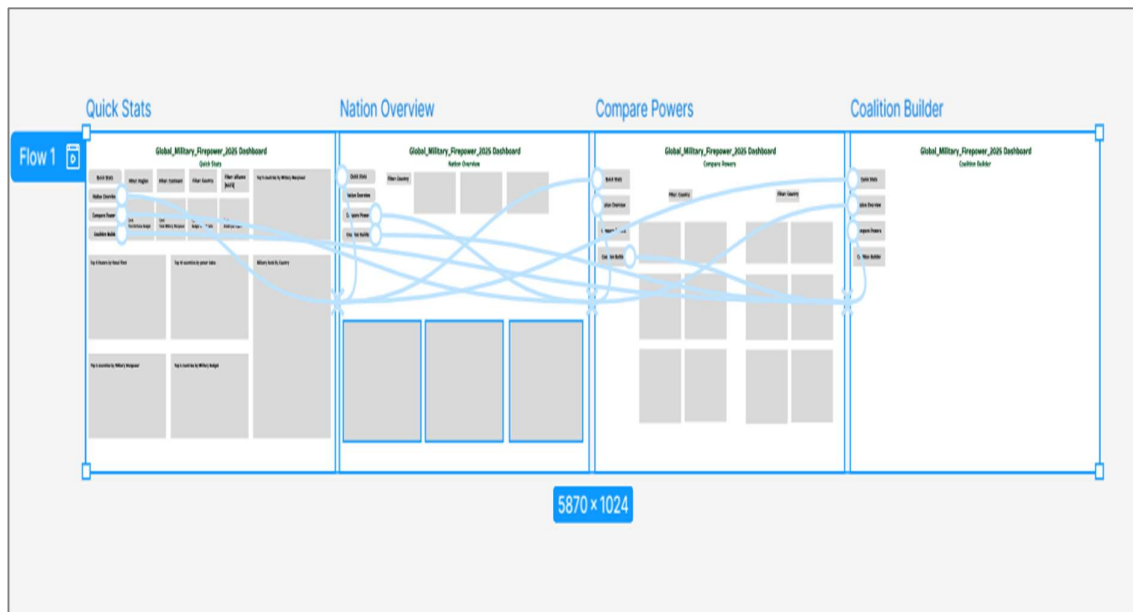


Figure 12. Overall flow of the dashboard planned using wireframe.

4.3.2 Power Bi

- Create global_power_2025.pbix
- First import the military_final.csv.
- Create 4 blank dashboards:
- Quick stats, Nation Overview, Compare Powers Coalition Builder.
- Insert Page Navigator Buttons in all 4 dashboards. We can alt+click to test if it navigates correctly.
- Next, sample data is plotted in the first dashboard Quick Stats, to check if it working properly.
- In quick stats, using slicer, filters are added. Filters: Region, Continent, Country and Alliance.
- For showing important KPI's cards are used. In cards, Total Defense Budget (USD) , Total Military Manpower, Budget to GDP Ratio and Assets per Capita is visualized

- For quick stats:
 - Top 5 countries by military power is visualized using Pie chart using columns country and total_manpower and applying filter Top N.
 - Top 5 Powers by Naval fleet is visualized using Donut Chart, using columns country and total_naval_fleet, and applying filter Top N.
 - Top 10 Countries by Power Index is visualized using Funnel where lower values is stronger country, using columns country and pwr_index, and in filters Top 10, but from bottom.

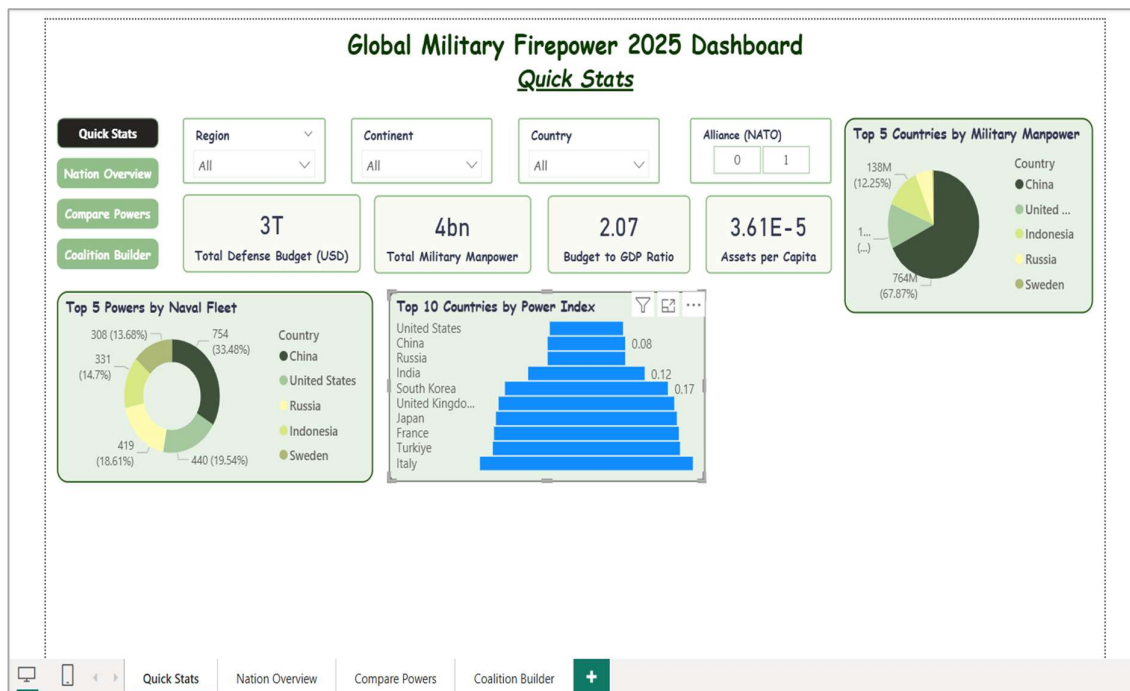


Figure 13. Quick Stats Dashboard (Operational)

Milestone 3: Full Dashboard Development

Module 5: Build Quick Stats and Nation Overview Dashboards

5.1 Quick Stats

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Quick Stats dashboard.

- Filters:
 - Region
 - Continent
 - Alliance

- Cards:
 - Total Country
 - Total Defense Budget
 - Total Military Manpower
 - Budget to GDP Ratio
 - Assets per capita

- Map:
 - Top 5 Countries by Total military manpower
 - Dimension: country
 - Measure: sum of military manpower

- Donut Chart:
 - Top 5 Powers by Active Personnel
 - Dimension: country
 - Measure: sum of Active Personnel

- Pie Chart:
 - Top 5 Countries by defense budget
 - Dimension: country
 - Measure: sum of defense budget
- Area Chart:
 - Top 10 countries by power index
 - Dimension: country
 - Measure: sum of power index

Use filter in everything for example:

country
top 5 by Sum of activ...

Filter type ⓘ

Top N ▼

Show items

Top ▼ 5

By value

Sum of active_person... ▼ ×

Apply filter

Figure 14: Top N filters

So in quick stats we can filter on the basis of nato alliance, continent and region and visualize the top countries on the basis of various metrics.

QUICK STATS DASHBOARD

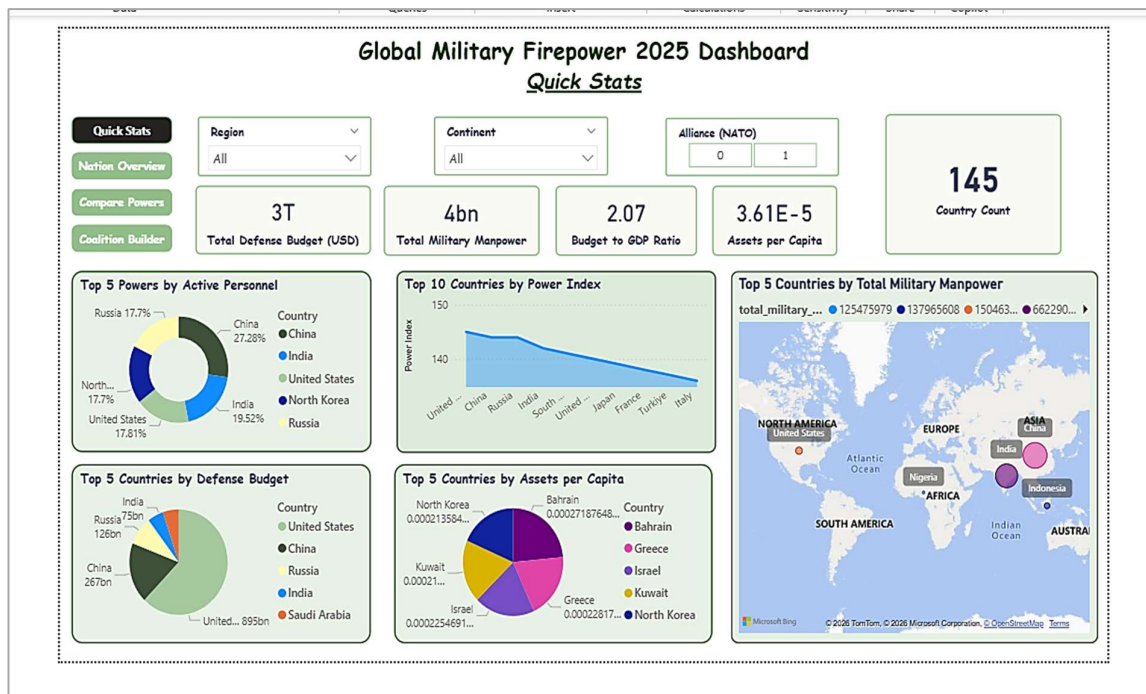


Figure 15: Quick Stats Dashboard

5.2 Nation Overview

Steps:

- Identify and list all KPIs required for the Nation Overview dashboard.
- Filters:
 - Country
 - Alliance

So in the Nation Overview dashboard, the visualizations can be filtered with the help of country filter, to know about each country.

- Cards:
 - Total land area (sq_km)
 - Total Population (sum)
 - Country: to show the selected country
 - Budget per soldier

- Budget to gdp ratio
- Multi row card: country, pwr index and power rank
- Map:
 - Country and Region
 - Dimension: Country
 - Measure: Region
- Bar Chart:
 - Defense Budget
 - Dimension: country
 - Measure: defense_budget_usd, country with highest budget
 - Using modelling, new measure:

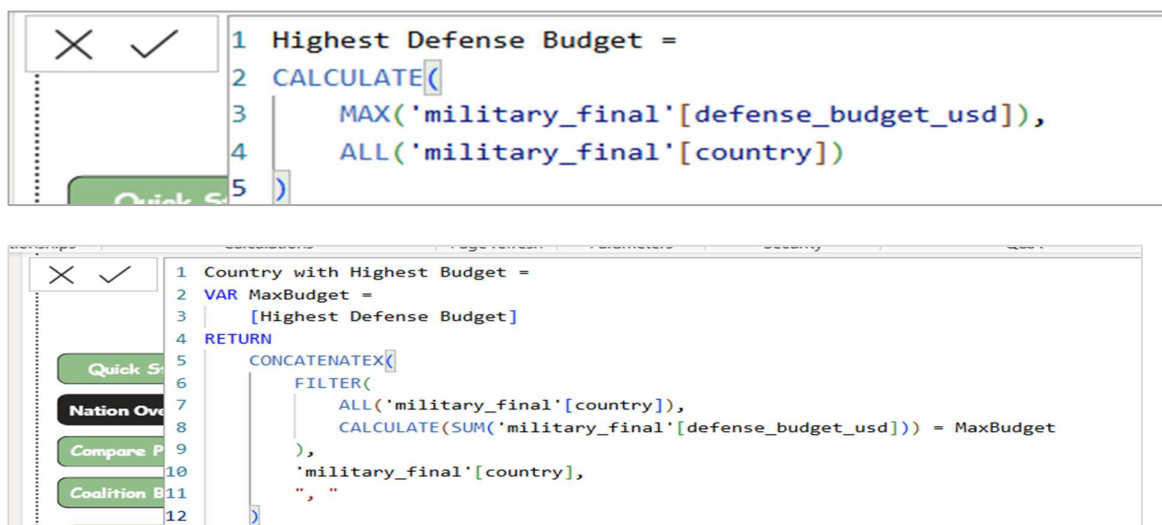


Figure 16: New Measure: Country with Highest Budget

- Pie Chart:
 - Active vs Reserve vs Paramilitary
 - Dimension: country
 - Measure: Paramilitary, Reserve_personnel, Active_personnel
 - Tooltips: Country and Power Rank
 - Naval, Tank, Aircraft
 - Dimension: country

- Measure: sum of Total_naval_fleet, tanks and Total aircraft
- Tooltips: country
- Gauge:
 - Military Participation Ratio
 - Sum of military_participation_ratio

NATION OVERVIEW DASHBOARD

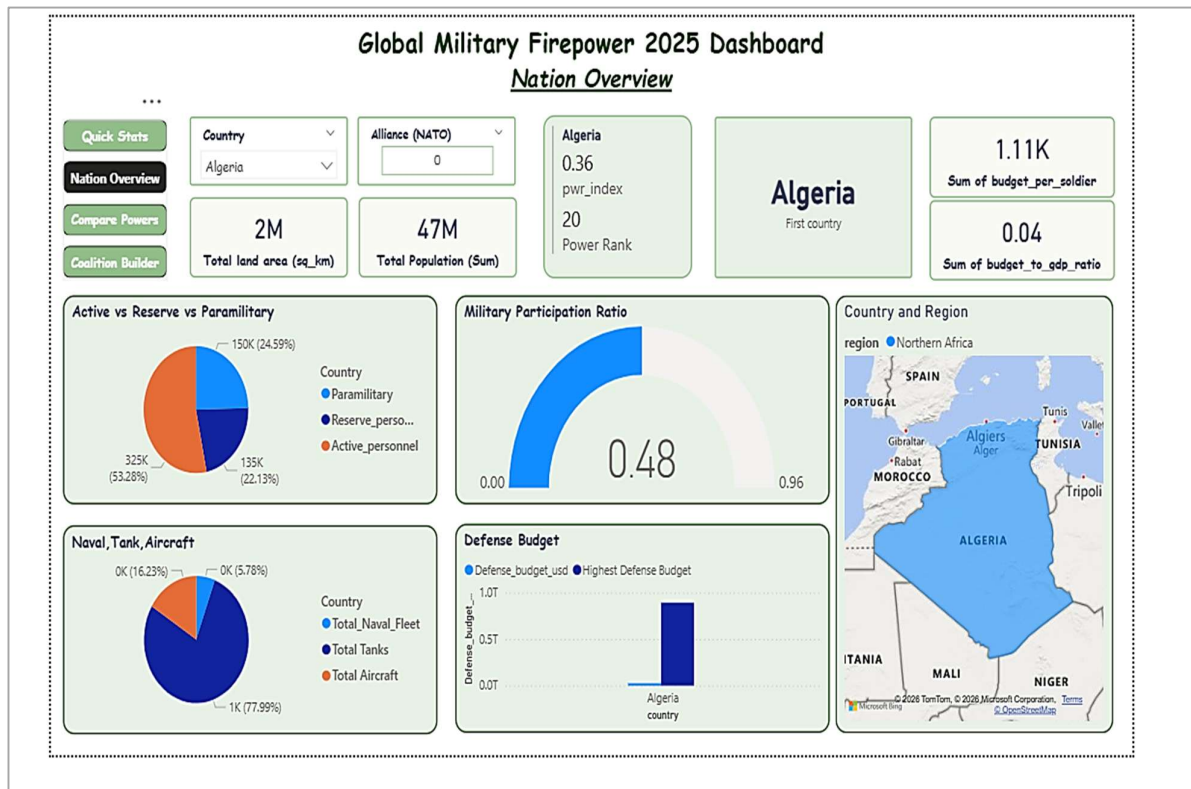


Figure 17: Nation Overview Dashboard

Module 6: Build Compare Powers and Coalition Builder Dashboards

6.1 Compare Powers

In this dashboard, two countries can be selected with the help of filters, and compare these countries on the basis of various power related metrics like, aircraft, tanks, naval etc.

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Compare Powers dashboard.
- Filters: Change the settings as select one.
 - Country A
 - Country B

For the selected country to filter the correct visual, select the filter visual, go to format, there edit interactions, there filter or none was selected according to requirements.

- Cards:
 - Country A selected
 - Country B selected
 - Total Military Manpower A
 - Total Military Manpower B
 - Budget to GDP Ratio A and B cards
 - Assets per capita A and B cards
 - Power index rank gap A and B cards.
- Bar chart:
 - i. Defense Budget (USD) Country A: country and defense_budget_usd
 - ii. Defense Budget (USD) Country B: country and defense_budget_usd
 - iii. Total Aircraft, Tanks and Naval Fleet by Country A

For total_aircrafts:

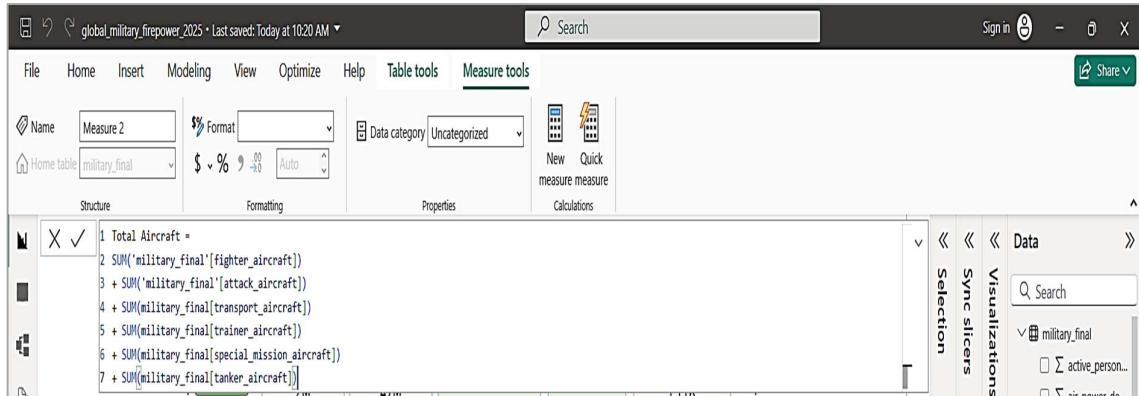


Figure 18: New measure: total_aircraft

- iv. Total Aircraft, Tanks and Naval Fleet by Country B
- v. Active Personnel: Country A
- vi. Active Personnel: Country B

Add data labels to all the charts.

COMPARE POWERS DASHBOARD

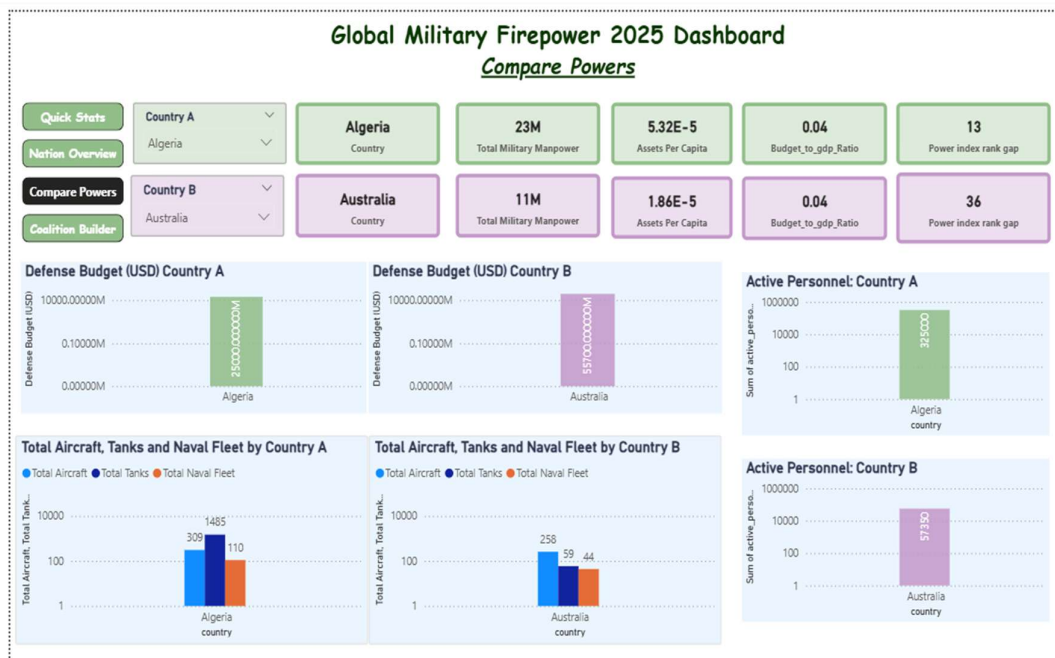


Figure 19: Compare Powers Dashboard

6.2 Coalition Builder

In this dashboard we can compare the power of the countries by combining two or three countries and comparing with other combination of countries, to find out which has better combined strength.

Steps:

- Load the final KPI dataset.
- Identify and verify Measures and Dimensions.
- Identify and list all KPIs required for the Coalition Builder.
- Filters: Change the settings as MULTIPLE SELECT.
 - Coalition A
 - Coalition B

For the selected country to filter the correct visual, select the filter visual, go to format, there edit interactions, there filter or none was selected according to requirements.

- Cards:
 - 2 Multiple row cards for selected countries.
 - Total Military Manpower for coalition A.
 - Total Military Manpower for coalition B.
 - Total Land area sq km for coalition A.
 - Total Land area sq km for coalition B
 - Total Assets per capita for coalition A.
 - Total Assets per capita for coalition B.
 - Total Defense Budget (USD) for coalition A.
 - Total Defense Budget (USD) for coalition B.
- Bar chart:
 - i. Total Naval Fleet (Coalition of countries A): sum_of_total_naval_fleet
 - ii. Total Naval Fleet (Coalition of countries B): sum_of_total_naval_fleet
 - iii. Total Aircraft (Coalition of countries A): sum_of_total_aircraft

- iv. Total Aircraft (Coalition of countries B): sum_of_total_aircraft
- v. Total Tanks (Coalition of countries A): sum_of_tanks
- vi. Total Tanks (Coalition of countries B): sum_of_tanks

COALITION BUILDER DASHBOARD

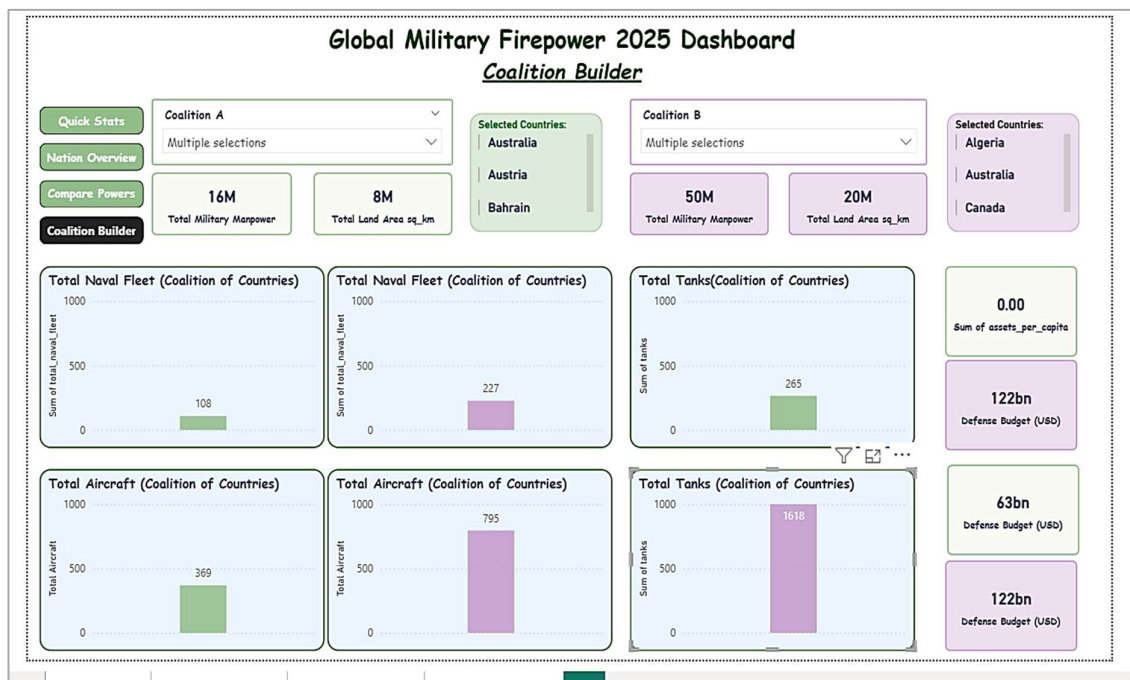


Figure 20: Coalition Builder Dashboard

Milestone 4: Final Review and Delivery

Module 7: Testing and Debugging

QA checklist:

i. Quick Stats Dashboard:

- Filters Testing
 - Region filter updates all charts correctly
 - Continent filter works properly
 - Alliance (NATO) filter shows correct countries
 - Multiple filters together work without errors
- KPI Validation
 - Total Defense Budget matches dataset
 - Total Military Manpower is correctly summed
 - Budget to GDP Ratio calculated correctly
 - Assets per Capita calculated correctly
 - Country Count changes based on filters
- Charts & Visuals
 - Top 5 Active Personnel pie chart shows correct percentages
 - Top 10 Power Index line chart sorts correctly
 - Defense Budget pie chart shows correct values
 - Assets per Capita chart displays correct countries
 - Map bubbles match manpower values
- Layout & UI
 - Titles are clear and correctly spelled
 - Axis labels are readable
 - No overlapping text
 - Tooltips show correct units (USD, bn, %, etc.)

ii. Nation Overview Dashboard:

- Filters Testing
 - Country dropdown updates all visuals
 - Alliance filter works correctly
 - Map highlights selected country
- KPI Validation
 - Total Land Area matches dataset
 - Total Population matches dataset
 - Power Rank matches source data
 - Budget per Soldier correct
 - Budget to GDP Ratio correct
- Charts & Visuals
 - Active vs Reserve vs Paramilitary pie chart correct
 - Military Participation Ratio gauge shows correct value
 - Defense Budget bar chart accurate
 - Naval, Tank, Aircraft chart accurate
- Layout & UI
 - Titles are clear and correctly spelled
 - Axis labels are readable
 - No overlapping text
 - Tooltips show correct units (USD, bn, %, etc.)

iii. Compare Powers Dashboard:

- Filter & Parameter Testing
 - Country A selector works
 - Country B selector works
 - Both countries can be changed independently
 - Charts update instantly when country changes

- KPI Validation
 - Manpower for Country A correct
 - Manpower for Country B correct
 - Assets per Capita correct for both
 - Budget to GDP ratio correct
 - Rank gap calculated correctly

- Charts & Visuals
 - Defense Budget charts show correct values
 - Active personnel charts accurate
 - Aircraft, Tank, Naval fleet comparison correct
 - Colors clearly differentiate Country A and B

- Layout & UI
 - Side-by-side comparison aligned properly
 - No overlapping bars
 - Labels readable

iv. **Coalition Builder Dashboard:**

- Filters Testing
 - Coalition A multi-select works
 - Coalition B multi-select works
 - Countries can be added/removed easily
 - Aggregated totals update correctly

- KPI Validation
 - Total Manpower = sum of selected countries
 - Total Land Area = correct sum
 - Total Naval Fleet = correct sum
 - Total Aircraft = correct sum
 - Total Tanks = correct sum
 - Total Defense Budget = correct sum

- Charts & Visuals
 - Coalition A and B shown separately
 - Bars represent total values correctly
 - No duplicate counting
 - Units clearly displayed

- Layout & UI
 - Coalition A and B clearly labelled
 - No visual clutter
 - Consistent color scheme
 - Dashboard loads without lag

All these navigations and filters are tested and working correctly as per the requirements.

Module 8: Documentation and GitHub Release

There are total 4 dashboards. Each dashboard can be accessed by ctrl+click on the navigation buttons.

i. Quick Stats

Purpose: Provides an overall summary of global military data.

How to Use:

- Use the filters at the top (Region, Continent, Alliance).
- KPI cards update automatically based on selected filters.
- Hover over charts to see detailed values.

ii. Nation Overview

Purpose: Shows detailed analysis for a selected country.

How to Use:

- Select a country from the Country dropdown.
- All KPIs and visuals update automatically.
- View:
 - Total land area
 - Population
 - Power rank
 - Budget per soldier
 - Military participation ratio
- Charts display personnel distribution, defense budget, and equipment data.

iii. Compare Powers

Purpose: Compare two countries side by side.

How to Use:

- Select Country A.
- Select Country B.
- The dashboard compares:
 - Military manpower
 - Defense budget
 - Assets per capita
 - Tanks, aircraft, and naval fleet
 - Power rank gap
- Charts update instantly when selections change.

iv. Coalition Builder

Purpose: Compare two groups of countries.

How to Use:

- Select multiple countries for Coalition A.
- Select multiple countries for Coalition B.
- The dashboard calculates total:
 - Manpower
 - Land area
 - Tanks
 - Aircraft
 - Naval fleet
 - Defense budget

- All totals are automatically aggregated.

Finally published to Power Bi online

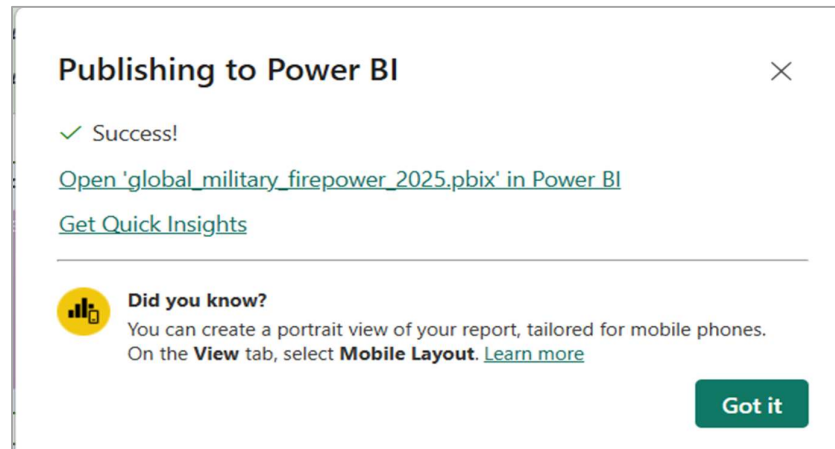


Figure 21: Publishing Dashboard to Power BI online.

Push all reports, data, workbooks and dashboard final version to GitHub.

This branch is 15 commits ahead of and 26 commits behind main .			Contribute
Bhagyabijay	Milestone 4 report	67e56a8 · 27 minutes ago	🕒 25 Commits
📁 dashboard	final dashboard		yesterday
📁 data	military_final.csv for milestone 2		last month
📁 docs	Milestone 4 report		27 minutes ago
📁 scripts	generate_kpis script		last month
📄 Figma_Wireframe_Navigation_Plan.pdf	Figma_Wireframe_Navigation_Plan		last month
📄 README.md	Update Data Scraping module status in README		last month
📄 links for military data.txt	Add files via upload		last month

Figure 22: Snapshot of Git Hub Repo.

Conclusion

Therefore, the project is completed by covering all the 4 milestone. The first milestone was focused on data scraping where around 145 countries data where extracted and cleaning of the data was done. In milestone 2, KPI engineering was done where more than five key performance indicators were accurately calculated and properly formatted to support meaningful analysis. The third milestone involved the global military dashboard development, where four interactive dashboards were built and connected to provide clear visualization, smooth navigation, and better user understanding. Finally, in the fourth milestone, delivery and quality assurance were performed to ensure full integration, correct functionality, and error-free performance.